Do EMT cells sit in a simplex, and are hybrid cells generalists?

An explainer of the archetype (Pareto-simplex) analysis across five single-cell EMT datasets. Thesis project, IISc · 22 October 2026. Every number below is read from `main_fits.csv` and `corner_annotations.csv` in `docs/emt_single_cell/best_simplex/`.

The answers in short

1. **Is there a significant simplex, and are its corners tasks?** A simplex is often significant: 26 of 38 main fits on the entire gene set, 16 of 38 on the KS EMT genes and 19 of 38 on the Hallmark EMT genes. But it means different things at the two levels. **Pooled** over samples, the corners are mostly the samples themselves (53% of corners of significant pooled simplices are ≥ 70% one cell line, clone, library, sample, tumour); Cook's pooled simplex is nothing but its four cell lines. **Within a single unit**, the corners are biological programmes (cycling, MYC/growth, a differentiated / KRAS-down epithelial state, metabolic, EMT / matrix), not the epithelial / hybrid / mesenchymal states. The EMT gene sets (KS, Hallmark) give fewer significant simplices than the entire gene set, and Hallmark behaves like KS.
2. **Are hybrid cells generalists, and more so than other cells?** Most cells are generalists, hybrid or not: under rule 1 a median 62% of hybrid cells and 56% of other cells are generalists (rule 2: 86% vs 81%; rule 3 about 100% for everyone). Hybrids are slightly more often generalists than other cells; median difference under rule 1 per dataset: Mouse SCC +10, MCF10A/HuMEC +6, Mouse PDAC +5, 4T1 clones +2, Cook +0 points. In no dataset do hybrid cells form a corner of their own.

1. What we wanted to know, and why

The idea (Pareto task inference). If a cell must perform several tasks and cannot be best at all of them at once, the best compromises lie inside a simple shape in gene-expression space: a line for two tasks, a triangle for three, a tetrahedron for four. Cells that specialise in one task sit at a corner (an *archetype*); cells that balance several sit inside (*generalists*). Finding such a shape, and naming its corners, is a way to read the tasks off the data.

Why EMT. Epithelial-to-mesenchymal transition is usually described as states (epithelial, hybrid, mesenchymal). Two competing pictures: (a) EMT states are task corners, perhaps with hybrid cells as their own corner; or (b) EMT is a continuum and hybrids are mixtures in the interior. The simplex view separates these: in (a) hybrids are specialists of a corner, in (b) they are generalists.

The two questions. (1) Is there a significant simplex, and do its corners correspond to tasks, when the fit uses A. the entire gene set, B. the KS EMT genes or C. the Hallmark EMT genes, both pooled over each dataset and within each tumour / cell line / sample? (2) How many hybrid cells are generalists, and how many generalists are hybrid, under three specialist rules?

Why three gene sets. The entire gene set lets any programme shape the cloud (unbiased but dominated by the biggest sources of variation, such as cell cycle or sample identity). The KS genes (Tan et al. 2014; the EMT scoring panel used in the earlier breast work) and the Hallmark EMT set (MSigDB, 200 genes) restrict the fit to EMT genes, so a simplex there would be an EMT-specific trade-off. Two EMT sets guard against conclusions that depend on one panel.

Why pooled and within. Pooling uses all cells but mixes samples, which differ for reasons unrelated to EMT (patient or mouse, cell line, technology). A pooled simplex can therefore be just sample identity. Fitting inside a single unit removes between-sample differences, so a within-unit simplex reflects variation among cells of one tumour or line.

2. The five datasets

dataset	what it is	units fitted (main)	hybrid cells	main arm (why)
Mouse skin SCC (GSE288137)	12 LKPR tumours; 8 scRNA, 4 multiome (nuclei)	pooled scRNA cells; 8 scRNA tumours	authors' Early-Hybrid + Hybrid	scRNA only: the multiome nuclei create a technical corner; multiome tumours kept in an appendix
Cook (GSE147405)	4 human lines (A549, DU145, MCF7, OVCA420) given TGF-β, sampled over time	pooled 4 lines; each line	middle two quartiles of the KS EMT score (no state labels)	cleaned: ≥ 10,000 UMI, mito ≤ 0.07, MALAT1/NEAT1/haemoglobin dropped
Mouse PDAC (GSE173958)	tumour cells of 2 mice at several sites	pooled in vivo; 7 libraries, mouse 1, mouse 2	KS-score bins Early- + Late-Hybrid	cleaned: ≥ 10,000 UMI, liver ambient-RNA genes dropped
4T1 clones (GSE303505)	5 clones + parental, one tumour each	pooled 6 clones; each clone	KS-score quartiles Early- + Late-Hybrid	cleaned: ≥ 10,000 UMI + immune / stromal contamination removed (clone E: light QC, 142 cells after cleaning)
MCF10A / HuMEC (GSE114687)	mammary cells in colonies; EMT rises toward the edge; Mock vs TGF-β	pooled 4 samples; 6 units (line, line × arm)	KS-score quartiles Early- + Late-Hybrid	cleaned: ≥ 10,000 UMI, MALAT1/NEAT1/haemoglobin dropped

Why one main fit per unit. Each dataset was fitted with a light-QC arm and a cleaned arm. The two disagree on whether a simplex is significant in many units (Cook 2 of 12 unit × gene-set pairs, PDAC 8 of 27, 4T1 7 of 15, GSE114687 8 of 18), so choosing whichever arm gives the lower p would inflate significance. The cleaned arm is the one that removes each dataset's known technical problem (low-depth cells, ambient RNA, contaminating immune or stromal cells), so it is the main arm; for SCC the analogous choice is scRNA only. Everything below counts units, one main fit each.

3. Methods and code

3.1 Fitting a simplex

For each unit and gene set: (i) build the log-normalised expression matrix (counts per 10,000, log1p; genes in ≥ 1% of cells; mitochondrial and ribosomal genes dropped) — `01_build_anndata.py`; (ii) PCA on the unit's cells with genes centred but not scaled (scaling would give rare noisy genes the same pull as strong ones) — `02_pca.py` (pooled, balanced cap per sample so large samples do not decide the shape) or `08_within_unit.py` / `07_within_line_clean.py` / `08d_within_tumour_geneset.py` (within, capped at 4,000 cells); (iii) archetypal analysis (PCHA, 15 random starts, the largest-volume solution kept) on the first k−1 PCs, for k = 3 to 7 pooled (3 to 5 for the newest fits) and k = 3, 4 within — `03_fit_and_test.py`, engine `src/archetypes.py`.

3.2 Is it significant? (permutation test)

The t-ratio is the volume of the fitted simplex divided by the volume of the convex hull of the cells. Each PC is then shuffled independently (destroying any joint shape while keeping each axis' spread) and refitted, 200 times (100 for the Hallmark, cleaned-pooled and SCC scRNA-only KS / Hallmark pooled fits). p = (1 + number of shuffles at least as good) / (1 + shuffles). For each fit the k with the lowest p is reported (ties: smaller k; pooled fits ignore k beyond the explained-variance elbow, as in the earlier report). Significant = p ≤ 0.05. **Caveat:** a small p says the cloud is not a shapeless blob; samples that differ from each other also pass, which is why corners must be named before they are called tasks.

3.3 What is each corner? (naming)

For each corner, the 5% of cells nearest to it are compared with all other cells of the fit on the 50 MSigDB Hallmark gene sets (per-cell score = mean log-expression of the set; effect = difference of means in SD units). The top set names the programme, e.g. E2F Targets / G2-M = cycling, MYC Targets = growth, KRAS Signaling Dn = a differentiated / quiescent epithelial state. The share of S/G2M cells, the dominant EMT state, the dominant sample and the top marker genes support the name. **Identity rule:** in a pooled fit, a corner whose cells are ≥ 70% one sample is an *identity corner* (it marks a sample, not a task). For fits on the Hallmark EMT genes, the EMT set itself is excluded from naming (circular). — `corners.py`.

3.4 Specialists and generalists (three rules)

Every cell gets barycentric weights on the k corners (they sum to 1; negative weights of cells outside the simplex are clipped). A cell is a specialist of a corner, or a generalist, by three rules:

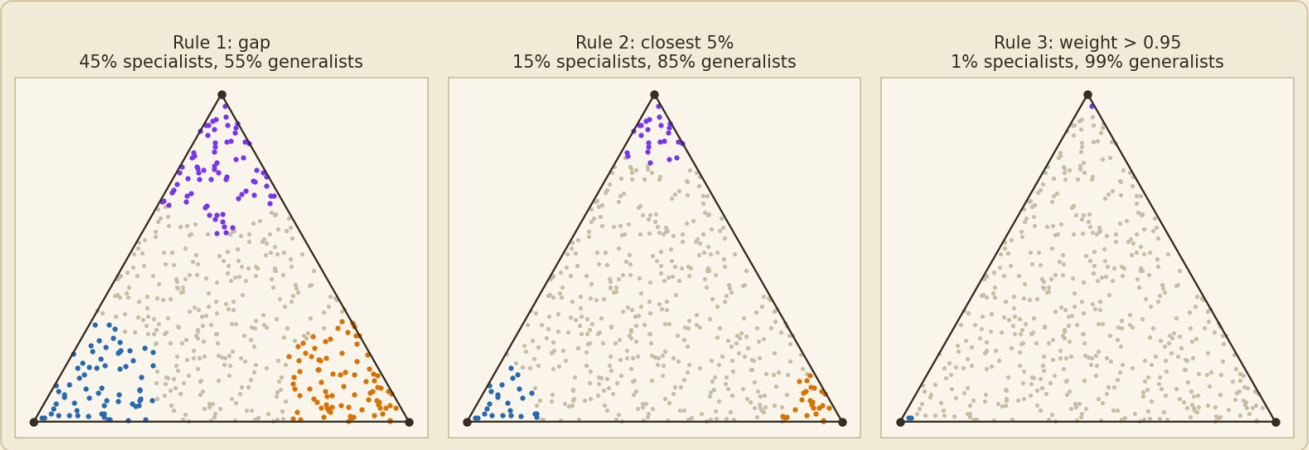


Figure 1. The three rules applied to 500 simulated cells inside a triangle. Coloured = specialists of a corner, grey = generalists.

- **Rule 1, purity gap** (used in the earlier Pal-breast work of this project): specialist if the largest weight exceeds the second largest by ≥ 0.35.
- **Rule 2, closest 5%** (Groves et al. 2022, main analysis): the 5% of cells nearest each corner are its specialists; everything else is a generalist.
- **Rule 3, weight > 0.95** (from the Groves 2022 methods; new to this project): specialist only if one weight exceeds 0.95. Because the fitted corners sit inside the data cloud, almost no cell qualifies, so this rule calls about 100% of cells generalist. It is the strictest and least informative rule.

The comparison. For each fit and rule we count what % of **hybrid cells** are generalists and what % of **all other cells** (epithelial + mesenchymal) are generalists. Same % = hybrids are not special; hybrids more often generalists = they sit in the interior more than other cells, as the picture of hybrids as mixtures predicts. The % of other cells comes from the overall generalist share and the hybrid numbers. — `analyze.py` (`rules()`), results in `main_fits.csv`.

4. Question 1: is there a simplex, and are its corners tasks?

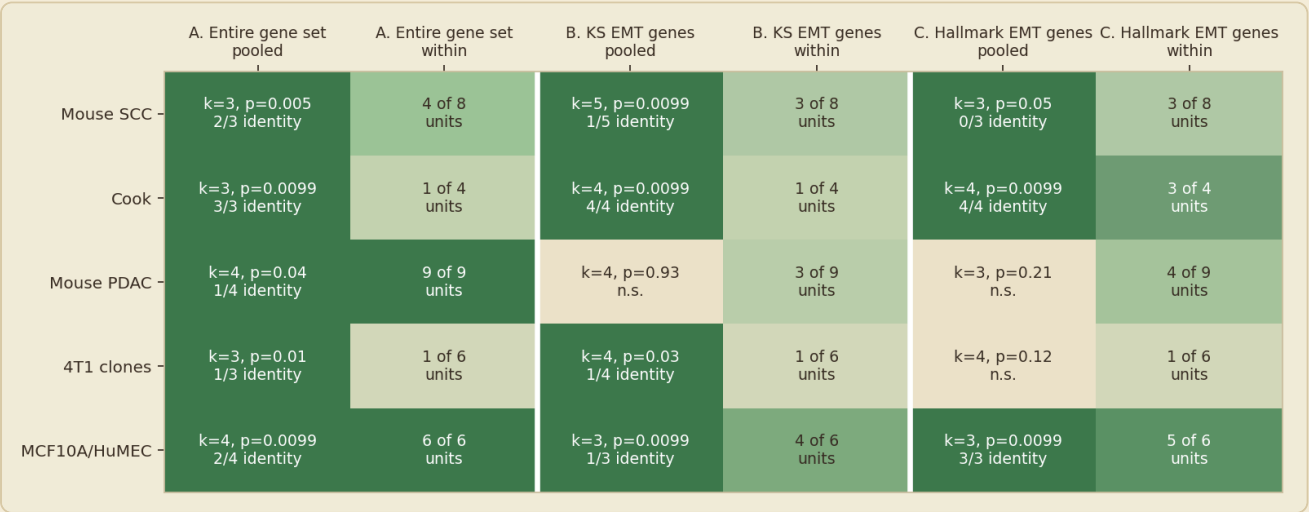


Figure 2. Overview. Pooled columns: best k and p of the pooled main fit (green = significant) and how many of its corners are identity corners (≥ 70% one sample). Within columns: number of units with a significant simplex (shade = fraction).

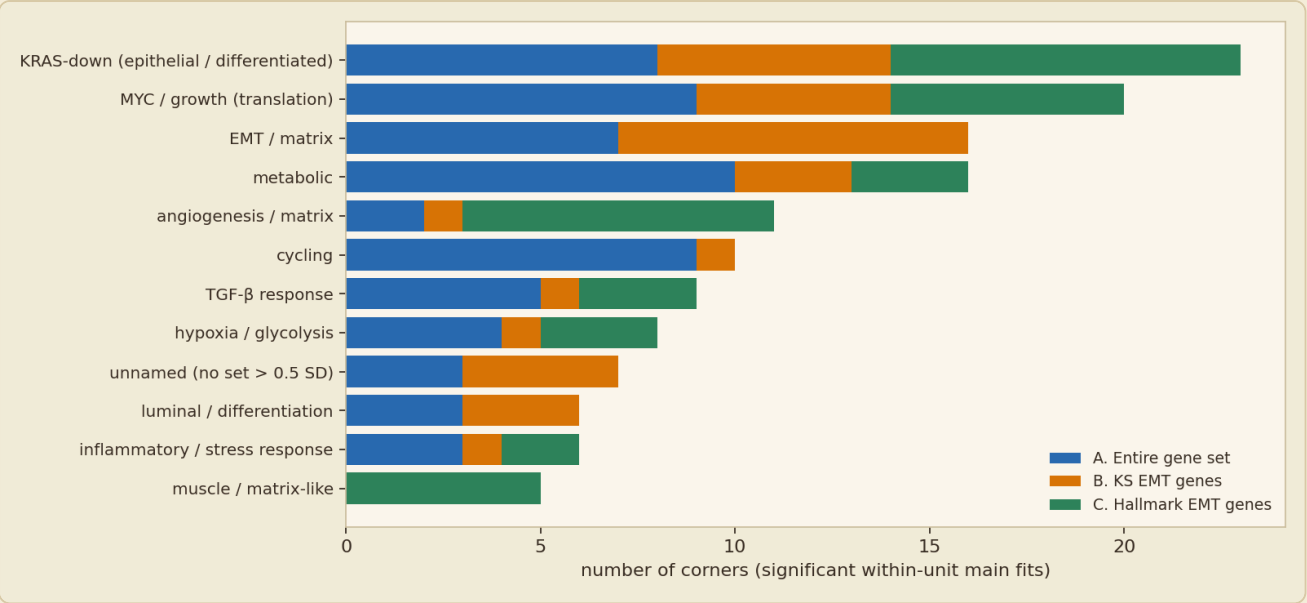


Figure 3. What the corners are, inside single units: names of all corners of significant within-unit main fits, by gene set.

4.1 Mouse SCC

Mouse skin squamous cell carcinoma: 12 real tumours (4 multiome, 8 scRNA), 23,959 cells, with the authors' four EMT states. Tumour, assay and EMT state are tangled.

Pooled.

- A. Entire gene set: significant (k=3, p=0.005); 2 of 3 corners are identity corners. Corners: V1 = SC15 (identity); V2 = SC2b1 (identity); V3 = MYC / growth (translation).
- B. KS EMT genes: significant (k=5, p=0.0099); 1 of 5 corners are identity corners. Corners: V1 = cycling; V2 = SC15 (identity); V3 = p53 / stress; V4 = EMT / matrix; V5 = KRAS-down (epithelial / differentiated).
- C. Hallmark EMT genes: significant (k=3, p=0.05); 0 of 3 corners are identity corners. Corners: V1 = angiogenesis / matrix; V2 = cycling; V3 = KRAS-down (epithelial / differentiated).

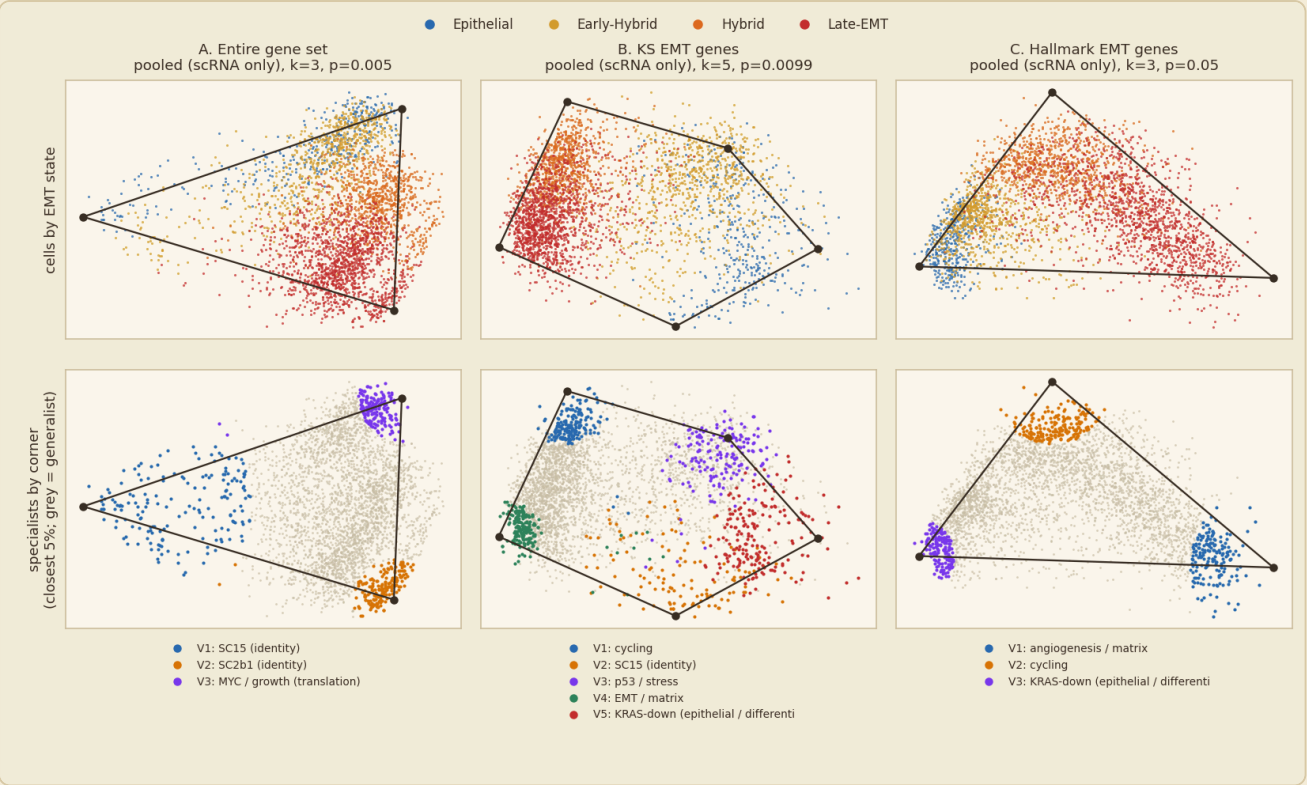


Figure 4. Mouse SCC, pooled main fits for the three gene sets. Top: cells coloured by EMT state. Bottom: the 5% of cells nearest each corner, coloured by corner and labelled (identity = ≥ 70% one tumour); grey = generalists. For k ≥ 4 the corners are projected on PC1/PC2.

Within each tumour.

- A. Entire gene set: 4 of 8 tumours have a significant simplex (SC15, SC32, SC2b1, SC2b3); most common corners: UV Response Dn (3), MYC / growth (translation) (3), EMT / matrix (2), inflammatory / stress response (1).
- B. KS EMT genes: 3 of 8 tumours have a significant simplex (SC32, SC2b3, SC35); most common corners: EMT / matrix (3), MYC / growth (translation) (2), p53 / stress (1), KRAS-down (epithelial / differentiated) (1).
- C. Hallmark EMT genes: 3 of 8 tumours have a significant simplex (SC15, SC32, SC2b1); most common corners: MYC / growth (translation) (2), angiogenesis / matrix (2), muscle / matrix-like (1), inflammatory / stress response (1).



Figure 5. Mouse SCC, every tumour: best k and p of the main fit per gene set (green = lower p, bold = significant) and, for significant fits, the corner names.

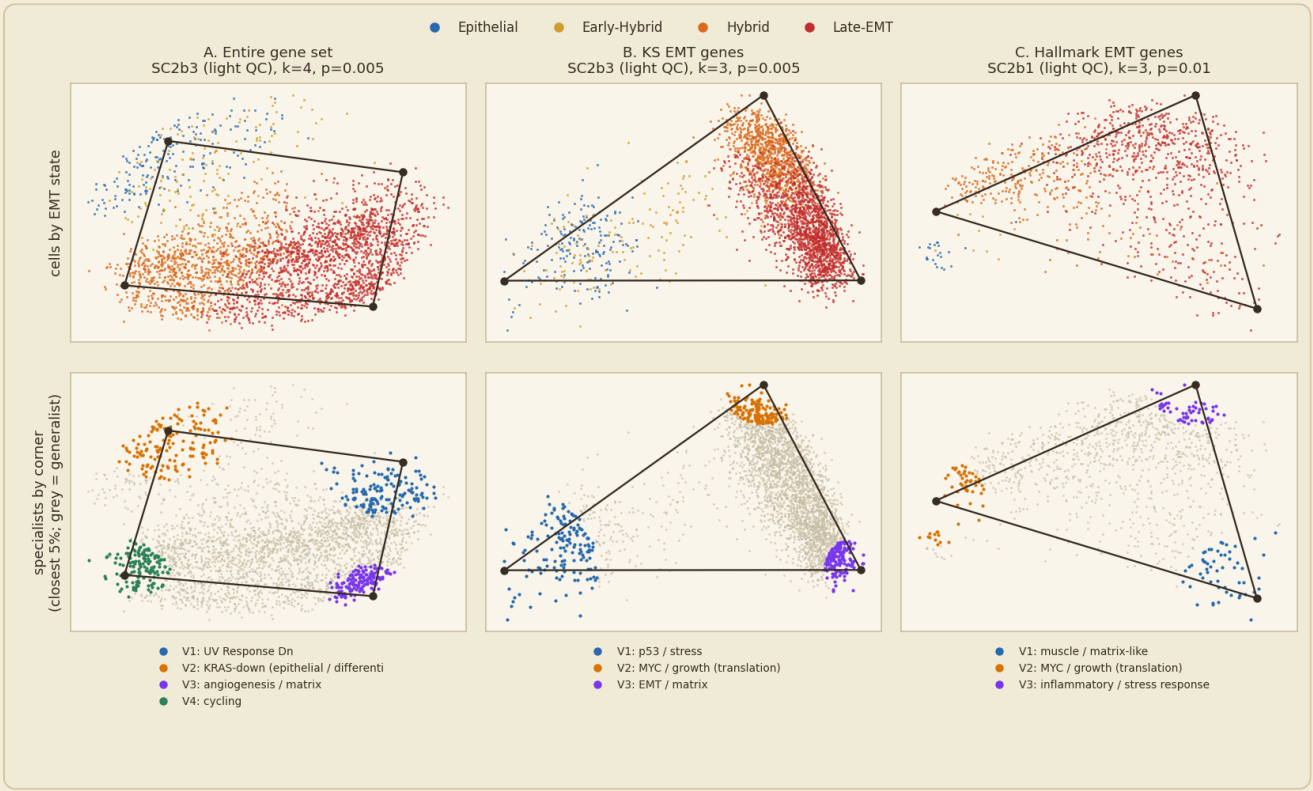


Figure 6. Mouse SCC, the most significant tumour for each gene set, drawn as in the pooled figure. Every other tumour is shown in the slide deck (`Explainer_simplex_results.pptx`).

4.2 Cook

Four human cancer cell lines given TGF-β and sampled over time. Time is an EMT coordinate independent of our score, but each line has its own identity.

Pooled.

- A. Entire gene set: significant (k=3, p=0.0099); 3 of 3 corners are identity corners. Corners: V1 = MCF7 (identity); V2 = A549 (identity); V3 = DU145 (identity).
- B. KS EMT genes: significant (k=4, p=0.0099); 4 of 4 corners are identity corners. Corners: V1 = OVCA420 (identity); V2 = A549 (identity); V3 = MCF7 (identity); V4 = DU145 (identity).
- C. Hallmark EMT genes: significant (k=4, p=0.0099); 4 of 4 corners are identity corners. Corners: V1 = DU145 (identity); V2 = OVCA420 (identity); V3 = A549 (identity); V4 = MCF7 (identity).

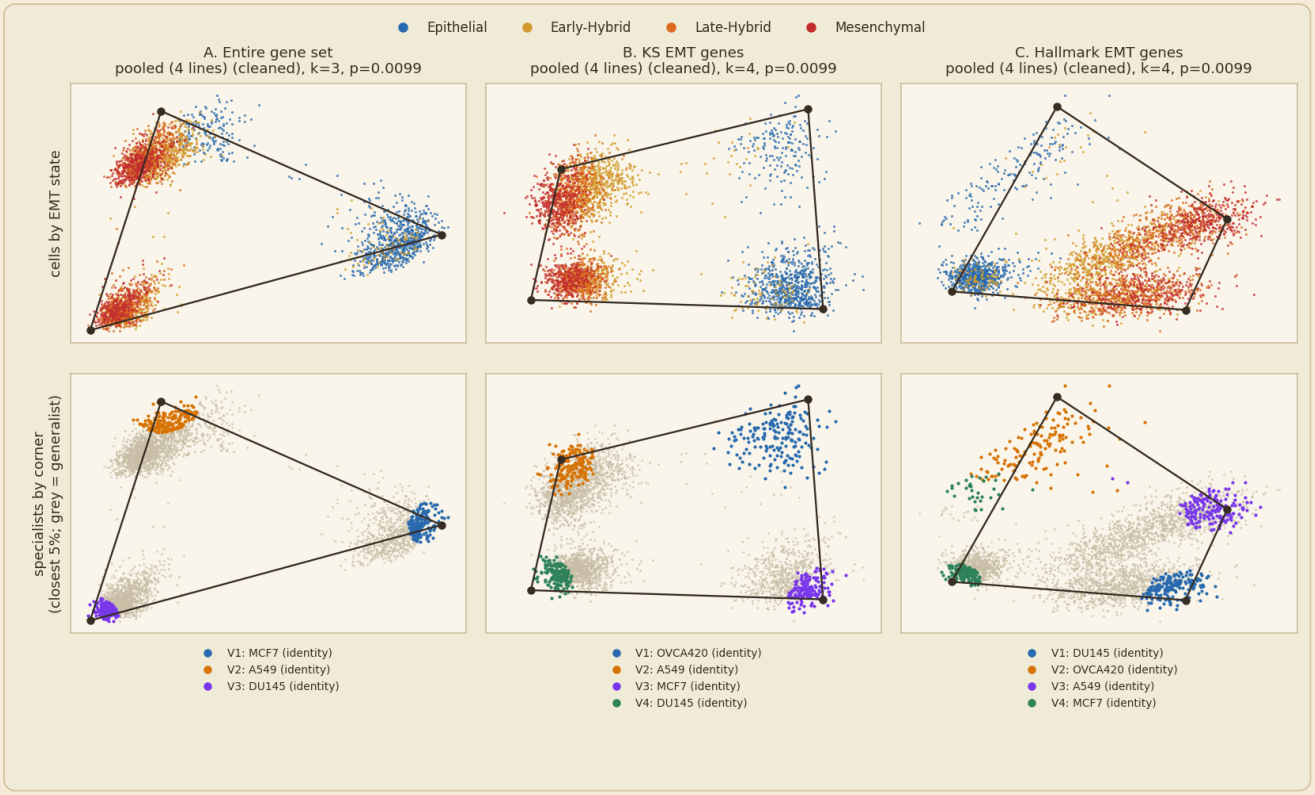


Figure 7. Cook, pooled main fits for the three gene sets. Top: cells coloured by EMT state. Bottom: the 5% of cells nearest each corner, coloured by corner and labelled (identity = ≥ 70% one cell line); grey = generalists. For k ≥ 4 the corners are projected on PC1/PC2.

Within each cell line.

- A. Entire gene set: 1 of 4 cell lines have a significant simplex (A549); most common corners: EMT / matrix (1), mTORC1 Signaling (1), luminal / differentiation (1).
- B. KS EMT genes: 1 of 4 cell lines have a significant simplex (MCF7); most common corners: inflammatory / stress response (1), MYC / growth (translation) (1), unnamed (1).
- C. Hallmark EMT genes: 3 of 4 cell lines have a significant simplex (A549, DU145, OVCA420); most common corners: TGF-β response (2), inflammatory / stress response (1), Fatty Acid Metabolism (1), muscle / matrix-like (1).

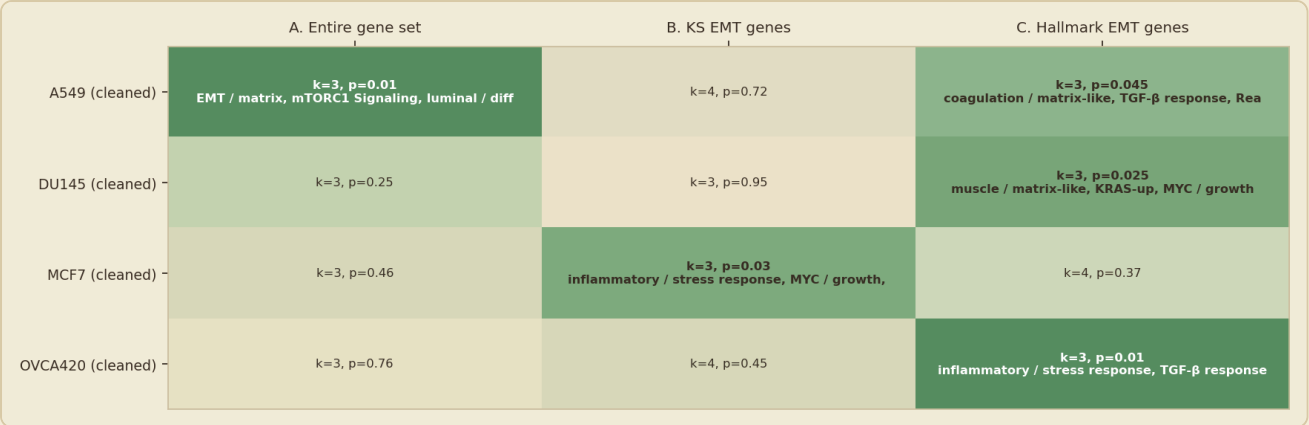


Figure 8. Cook, every cell line: best k and p of the main fit per gene set (green = lower p, bold = significant) and, for significant fits, the corner names.

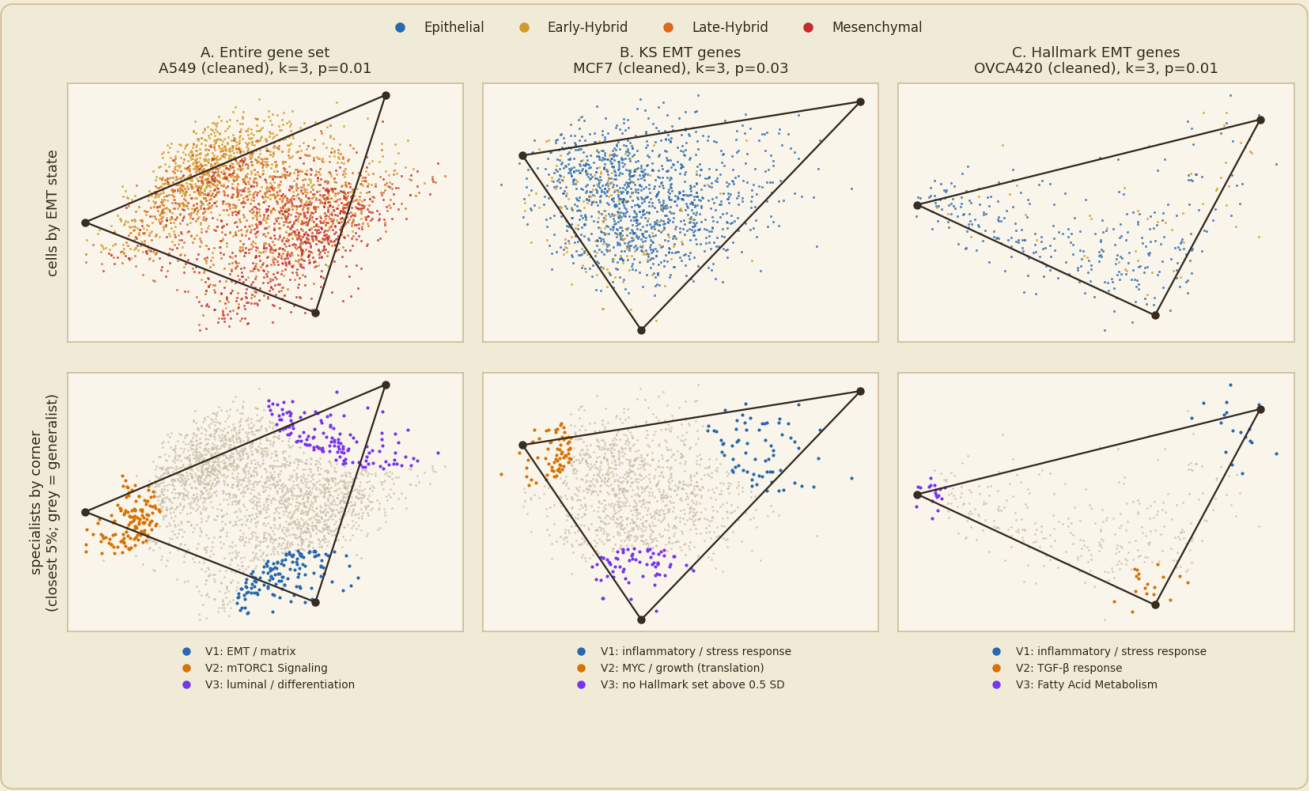


Figure 9. Cook, the most significant cell line for each gene set, drawn as in the pooled figure. Every other cell line is shown in the slide deck (`Explainer_simplex_results.pptx`).

4.3 Mouse PDAC

Mouse pancreatic tumour cells from two mice at several body sites (primary, liver, metastasis). EMT states are ours (KS-score bins), so partly circular.

Pooled.

- A. Entire gene set: significant (k=4, p=0.04); 1 of 4 corners are identity corners. Corners: V1 = luminal / differentiation; V2 = cycling; V3 = EMT / matrix; V4 = M2-Met (identity).
- B. KS EMT genes: no significant pooled simplex (k=4, p=0.93).
- C. Hallmark EMT genes: no significant pooled simplex (k=3, p=0.21).

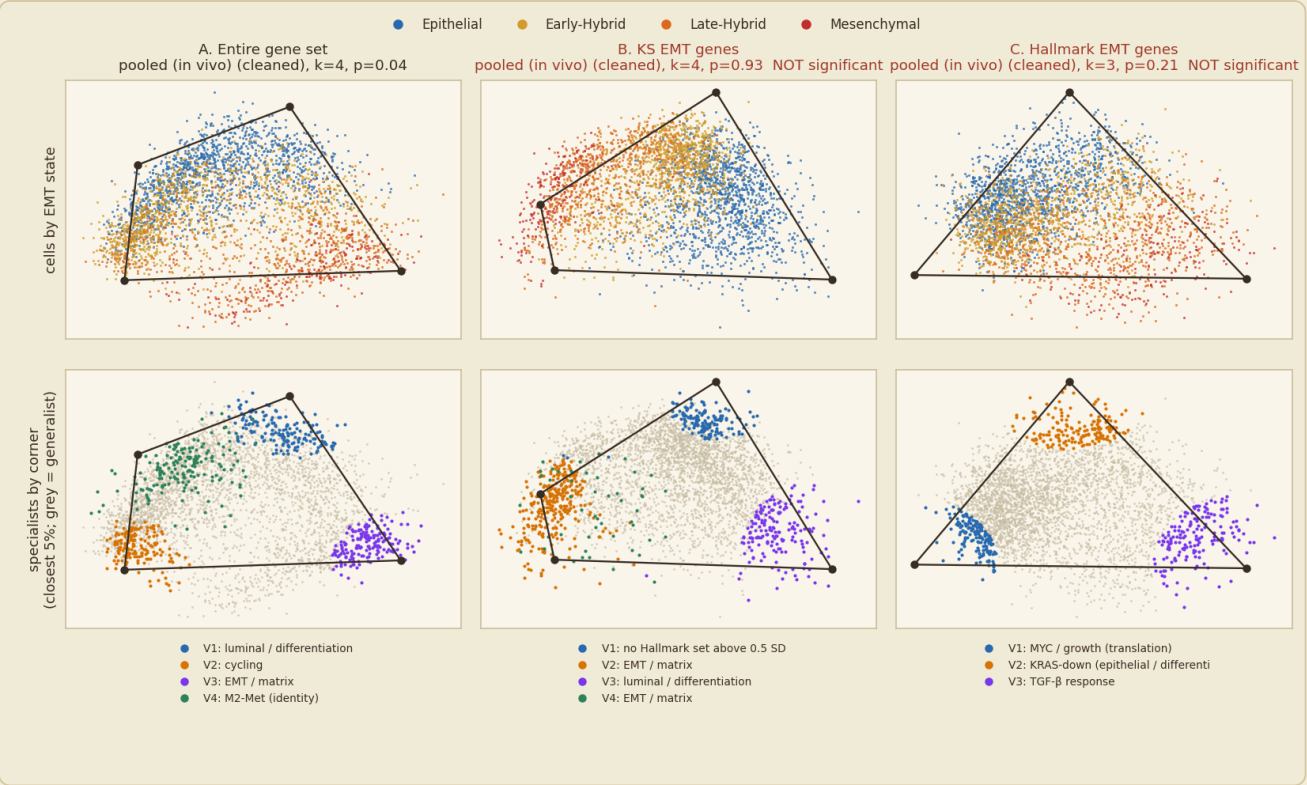


Figure 10. Mouse PDAC, pooled main fits for the three gene sets. Top: cells coloured by EMT state. Bottom: the 5% of cells nearest each corner, coloured by corner and labelled (identity = ≥ 70% one library); grey = generalists. For k ≥ 4 the corners are projected on PC1/PC2.

Within each library.

- A. Entire gene set: 9 of 9 librarys have a significant simplex (M1, M1-PT, M1-SS, M1-Liver, M1-Met, M2, M2-PT, M2-Liver, M2-Met); most common corners: metabolic (9), TGF-β response (4), MYC / growth (translation) (4), hypoxia / glycolysis (3).
- B. KS EMT genes: 3 of 9 librarys have a significant simplex (M1-SS, M2, M2-Met); most common corners: EMT / matrix (3), metabolic (2), luminal / differentiation (2), MYC / growth (translation) (2).
- C. Hallmark EMT genes: 4 of 9 librarys have a significant simplex (M1-SS, M1-Met, M2, M2-PT); most common corners: metabolic (3), angiogenesis / matrix (3), MYC / growth (translation) (2), KRAS-down (epithelial / differentiated) (2).

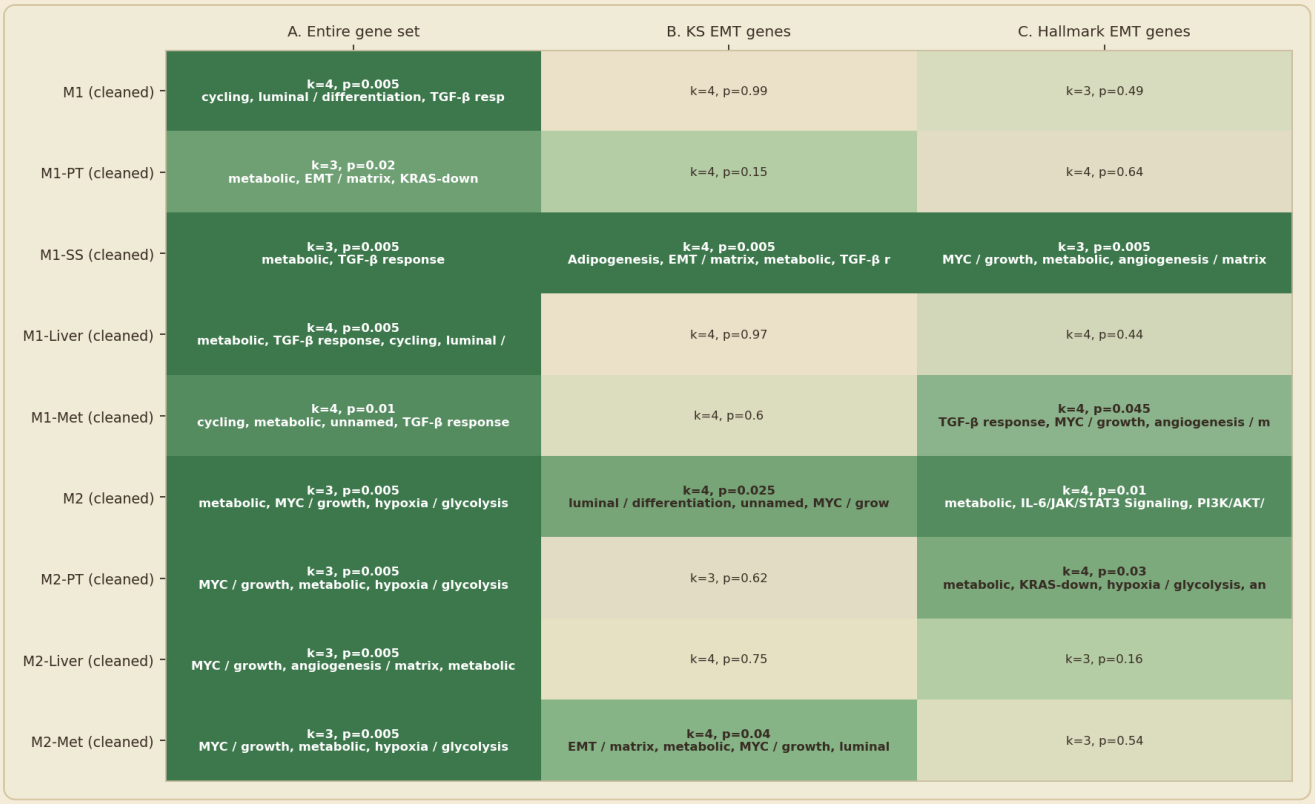


Figure 11. Mouse PDAC, every library: best k and p of the main fit per gene set (green = lower p, bold = significant) and, for significant fits, the corner names.

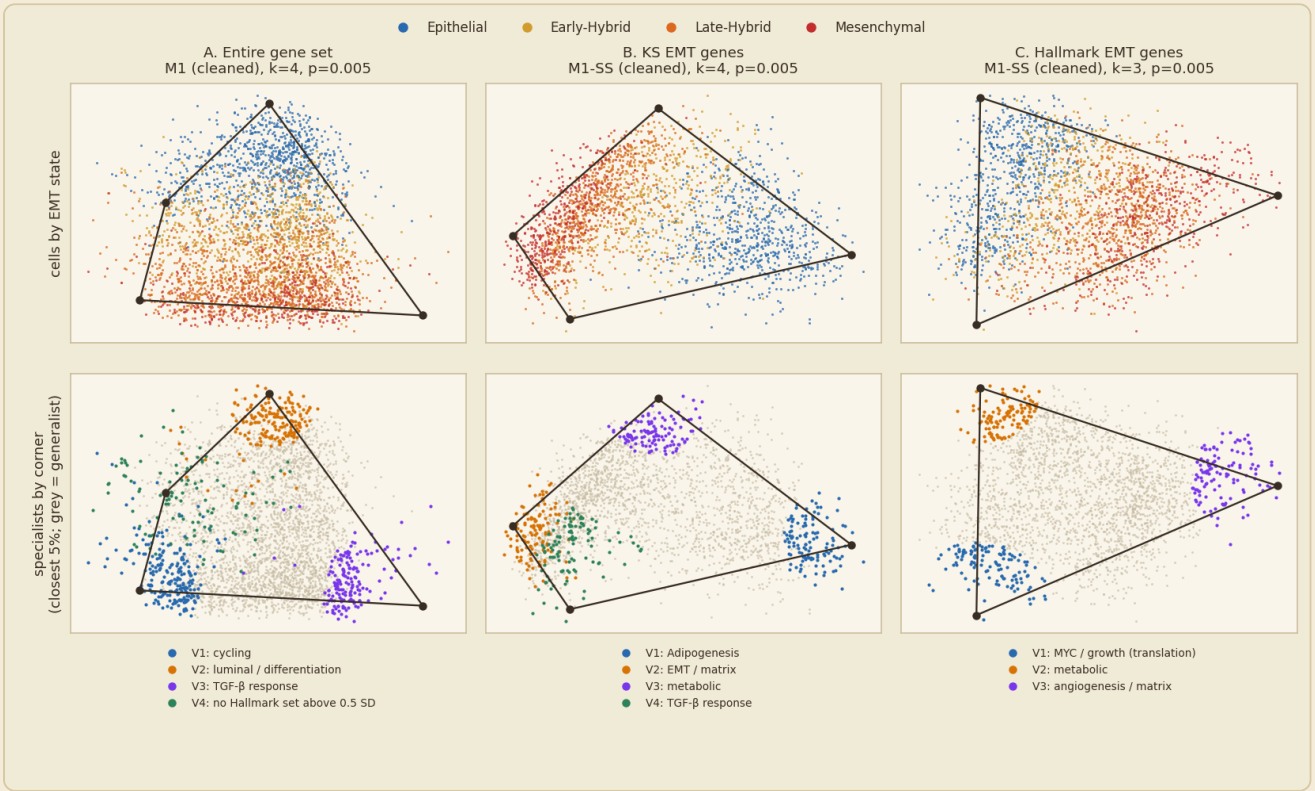


Figure 12. Mouse PDAC, the most significant library for each gene set, drawn as in the pooled figure. Every other library is shown in the slide deck (`Explainer_simplex_results.pptx`).

4.4 4T1 clones

Five single-cell clones of mouse 4T1 breast cancer chosen along an epithelial–mesenchymal spectrum, plus the parental line, one tumour each.

Pooled.

- A. Entire gene set: significant (k=3, p=0.01); 1 of 3 corners are identity corners. Corners: V1 = cycling; V2 = KRAS-up (squamous / differentiation); V3 = M (identity).
- B. KS EMT genes: significant (k=4, p=0.03); 1 of 4 corners are identity corners. Corners: V1 = TGF-β response; V2 = epithelial junction; V3 = no Hallmark set above 0.5 SD; V4 = M (identity).
- C. Hallmark EMT genes: no significant pooled simplex (k=4, p=0.12).

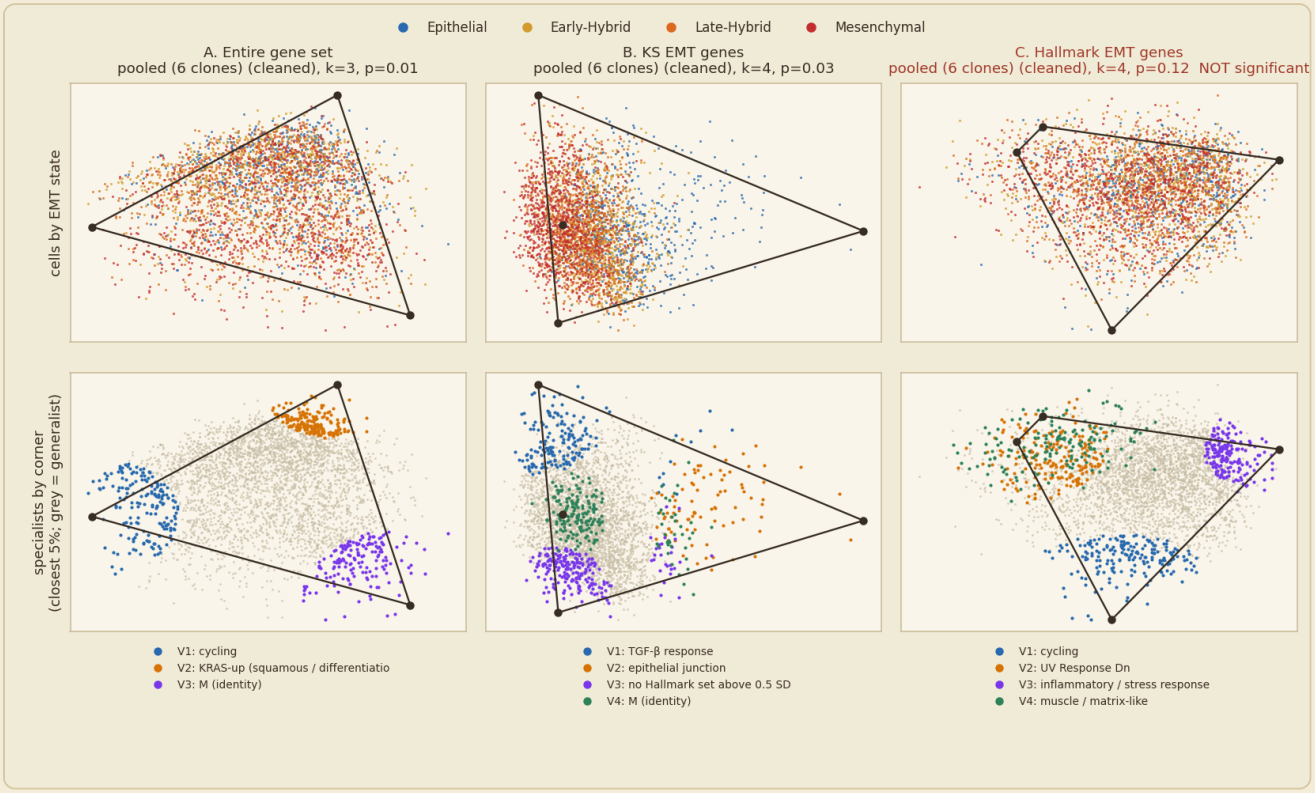


Figure 13. 4T1 clones, pooled main fits for the three gene sets. Top: cells coloured by EMT state. Bottom: the 5% of cells nearest each corner, coloured by corner and labelled (identity = ≥ 70% one clone); grey = generalists. For k ≥ 4 the corners are projected on PC1/PC2.

Within each clone.

- A. Entire gene set: 1 of 6 clones have a significant simplex (E); most common corners: unnamed (1), metabolic (1), cycling (1).
- B. KS EMT genes: 1 of 6 clones have a significant simplex (EM2); most common corners: luminal / differentiation (1), coagulation / matrix-like (1), Unfolded Protein Response (1).
- C. Hallmark EMT genes: 1 of 6 clones have a significant simplex (E); most common corners: KRAS-down (epithelial / differentiated) (1), p53 / stress (1), angiogenesis / matrix (1).

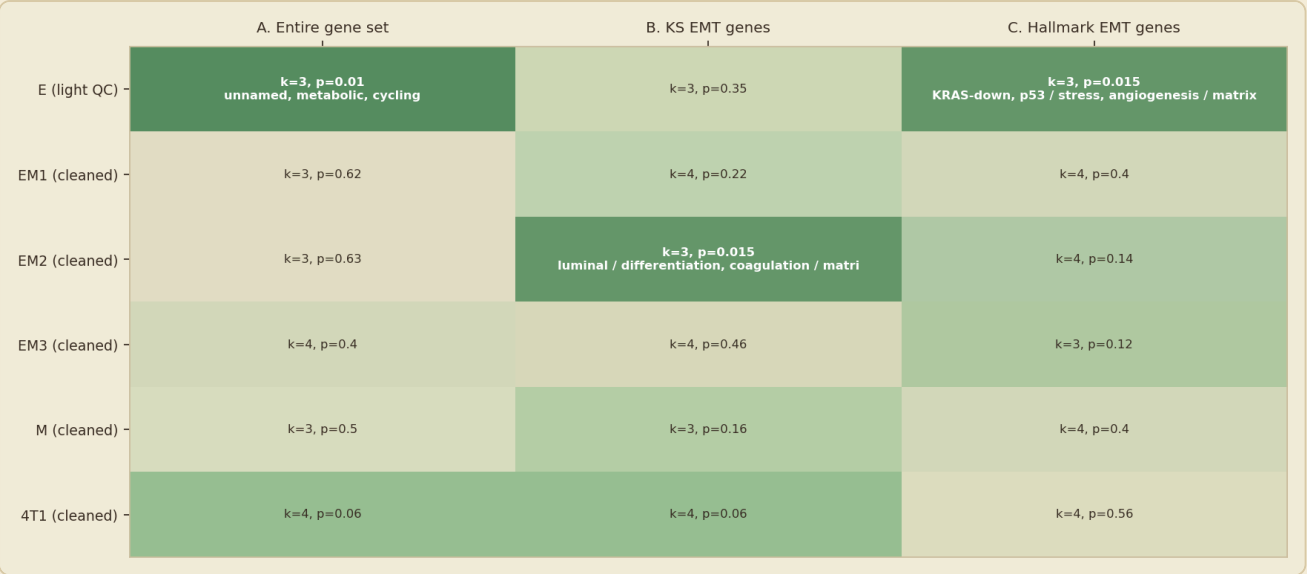


Figure 14. 4T1 clones, every clone: best k and p of the main fit per gene set (green = lower p, bold = significant) and, for significant fits, the corner names.

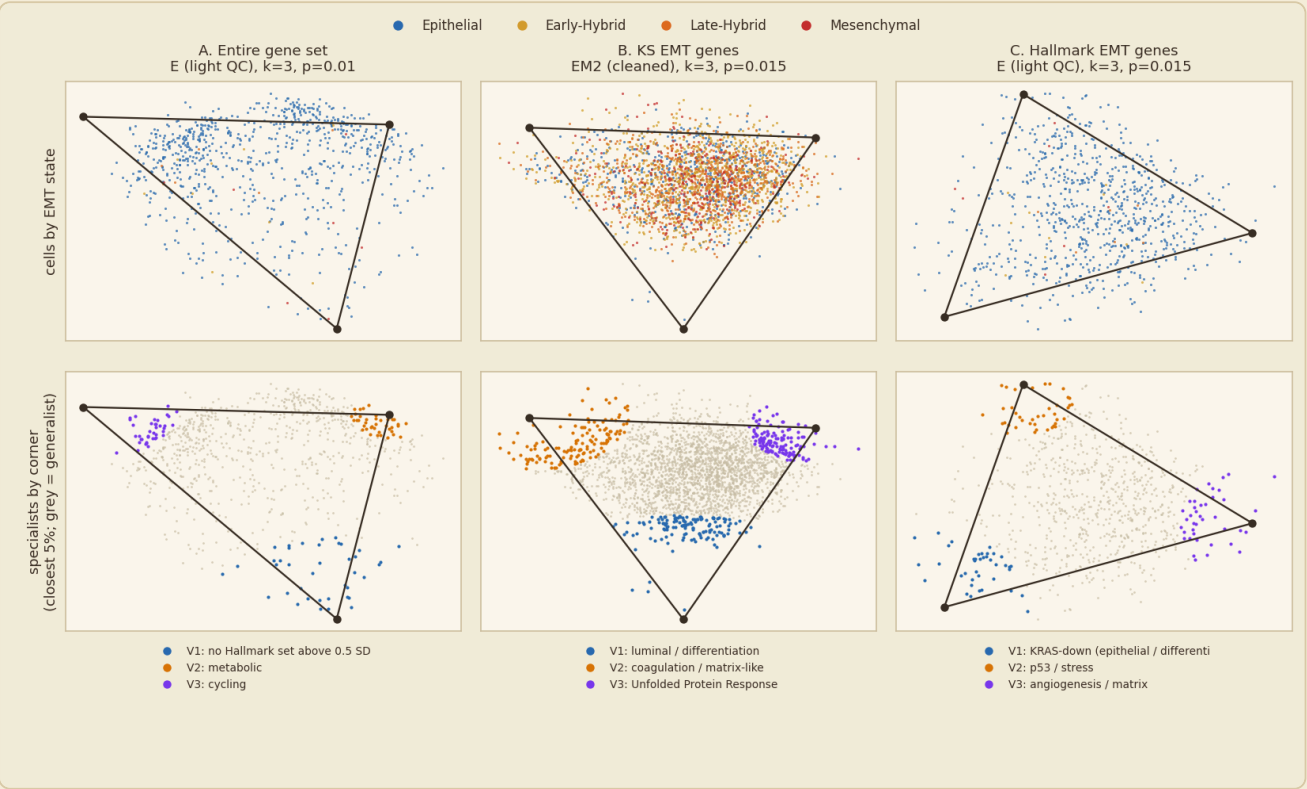


Figure 15. 4T1 clones, the most significant clone for each gene set, drawn as in the pooled figure. Every other clone is shown in the slide deck (`Explainer_simplex_results.pptx`).

4.5 MCF10A/HuMEC

Human mammary cells (MCF10A, HuMEC) grown as colonies so EMT rises with distance from the colony edge; Mock vs TGF-β. Induced EMT in culture, no replicates.

Pooled.

- A. Entire gene set: significant (k=4, p=0.0099); 2 of 4 corners are identity corners. Corners: V1 = MCF10A_TGFB (identity); V2 = KRAS-down (epithelial / differentiated); V3 = p53 / stress; V4 = MCF10A_Mock (identity).

- B. KS EMT genes: significant (k=3, p=0.0099); 1 of 3 corners are identity corners. Corners: V1 = p53 / stress; V2 = KRAS-down (epithelial / differentiated); V3 = MCF10A_TGFB (identity).
- C. Hallmark EMT genes: significant (k=3, p=0.0099); 3 of 3 corners are identity corners. Corners: V1 = MCF10A_Mock (identity); V2 = MCF10A_TGFB (identity); V3 = HuMEC_TGFB (identity).

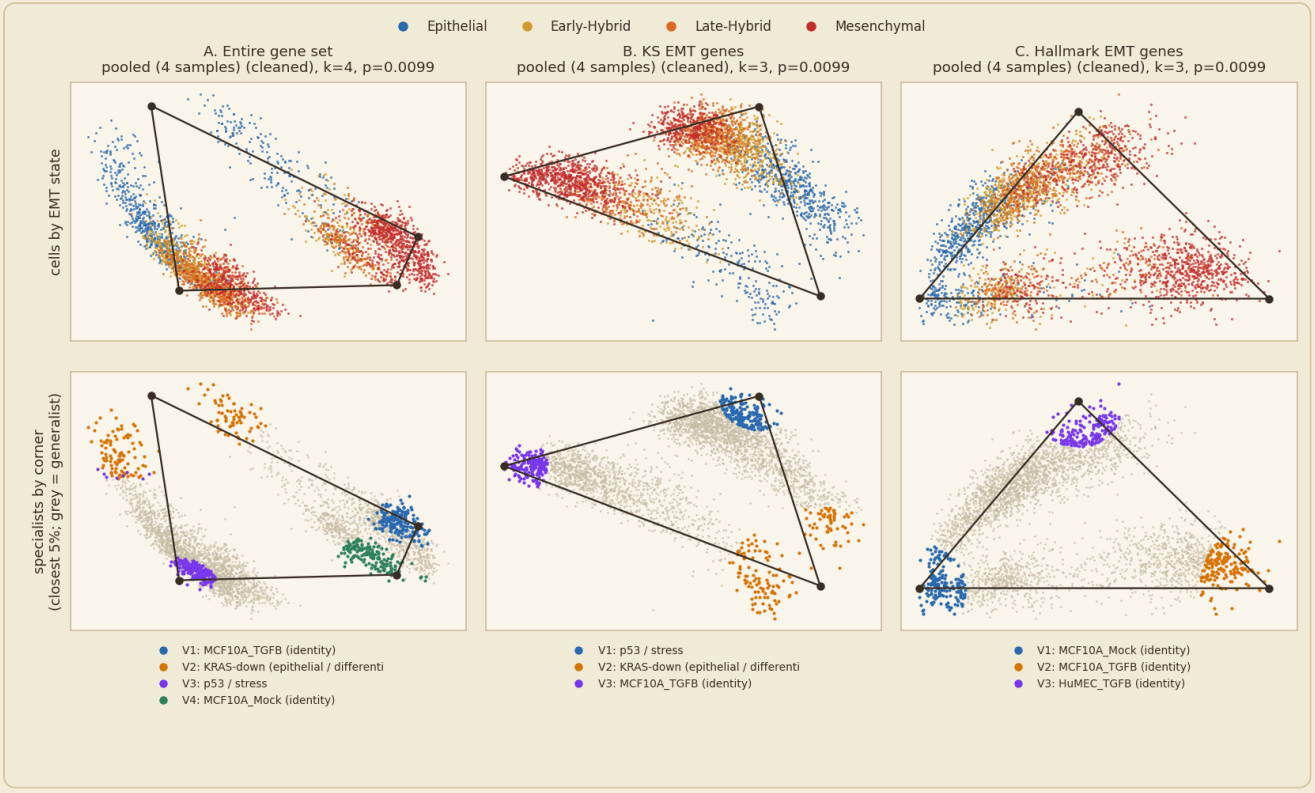


Figure 16. MCF10A/HuMEC, pooled main fits for the three gene sets. Top: cells coloured by EMT state. Bottom: the 5% of cells nearest each corner, coloured by corner and labelled (identity = ≥ 70% one sample); grey = generalists. For k ≥ 4 the corners are projected on PC1/PC2.

Within each sample.

- A. Entire gene set: 6 of 6 samples have a significant simplex (MCF10A, MCF10A_Mock, MCF10A_TGFB, HuMEC, HuMEC_Mock, HuMEC_TGFB); most common corners: KRAS-down (epithelial / differentiated) (6), cycling (4), EMT / matrix (3), MYC / growth (translation) (2).
- B. KS EMT genes: 4 of 6 samples have a significant simplex (MCF10A, MCF10A_Mock, HuMEC_Mock, HuMEC_TGFB); most common corners: KRAS-down (epithelial / differentiated) (5), EMT / matrix (3), hypoxia / glycolysis (1), unnamed (1).
- C. Hallmark EMT genes: 5 of 6 samples have a significant simplex (MCF10A, MCF10A_Mock, MCF10A_TGFB, HuMEC_Mock, HuMEC_TGFB); most common corners: KRAS-down (epithelial / differentiated) (5), muscle / matrix-like (2), coagulation / matrix-like (2), hypoxia / glycolysis (2).

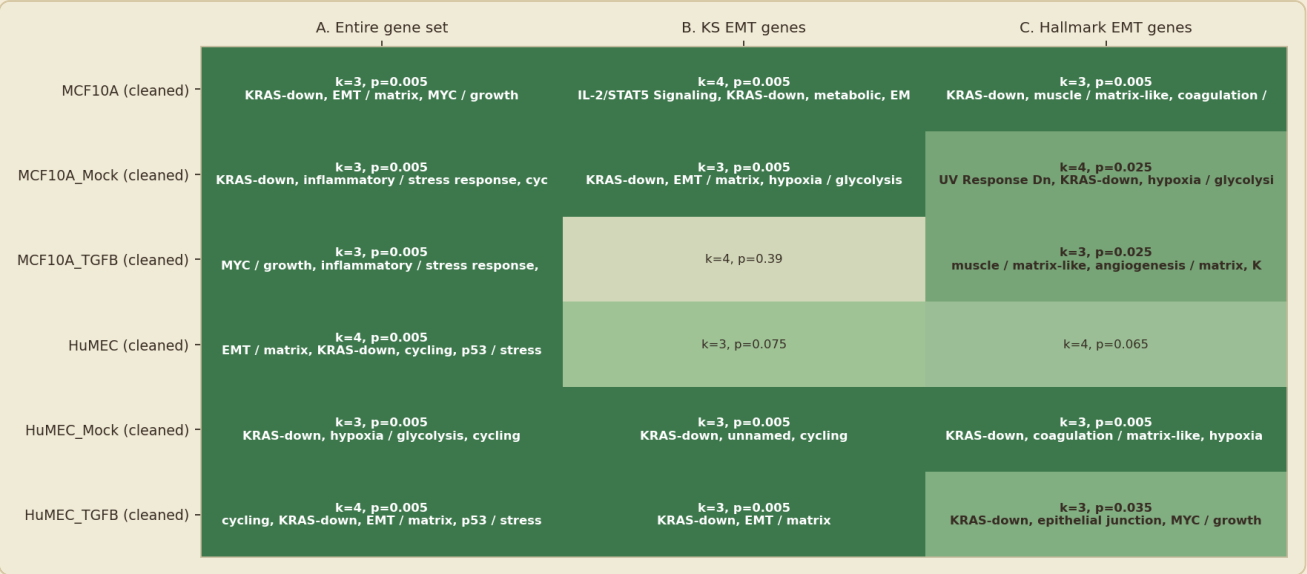


Figure 17. MCF10A/HuMEC, every sample: best k and p of the main fit per gene set (green = lower p, bold = significant) and, for significant fits, the corner names.

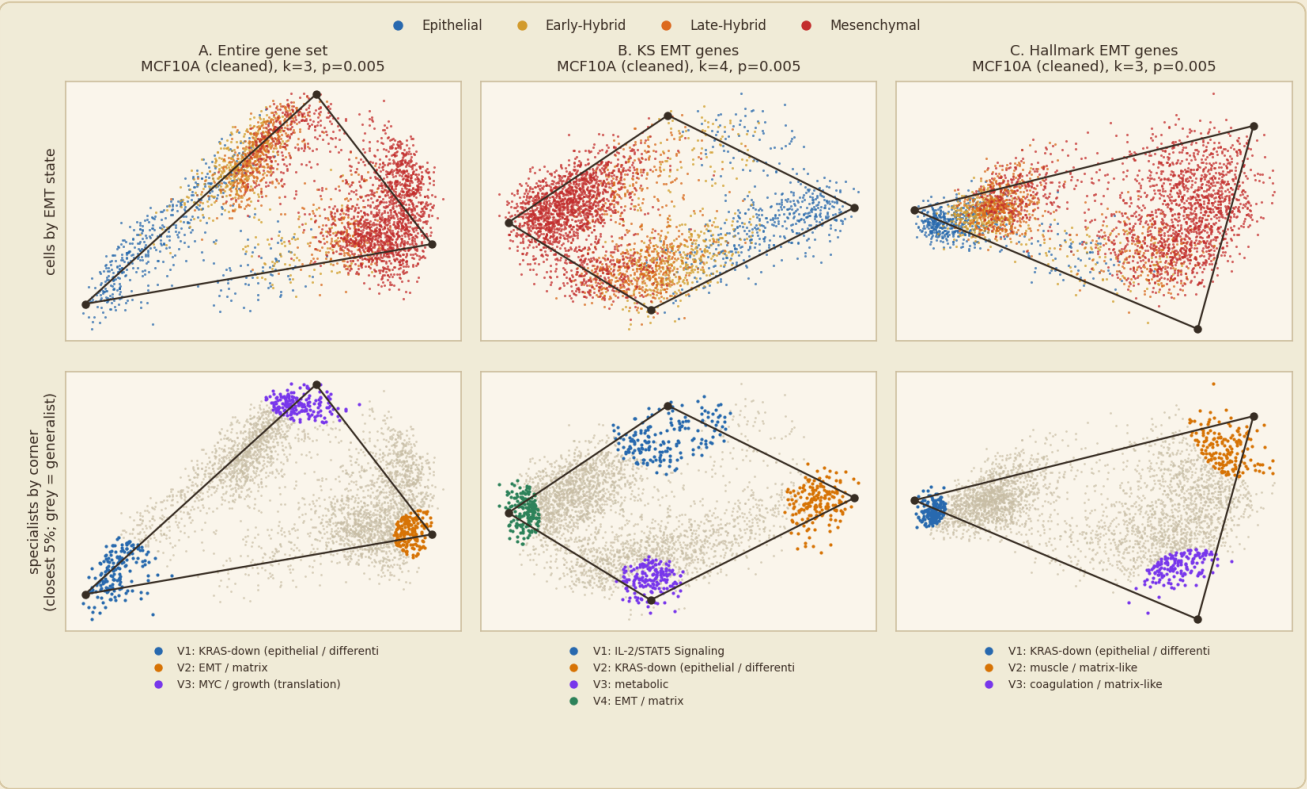


Figure 18. MCF10A/HuMEC, the most significant sample for each gene set, drawn as in the pooled figure. Every other sample is shown in the slide deck (`Explainer_simplex_results.pptx`).

4.6 What we saw (Question 1)

- **Gene set matters.** The entire gene set gives significant simplices most often (26 of 38 main fits), the EMT-only sets less often (KS 16, Hallmark 19 of 38). Hallmark does not rescue the EMT-only result: it behaves like KS.
- **Pooled shapes are mostly identity.** 53% of corners of significant pooled simplices are one sample. In Cook every pooled corner is a cell line; in MCF10A/HuMEC most are samples; in SCC two of three all-gene corners are tumours.

- **Inside units, the corners are programmes.** Cycling, MYC/growth, a differentiated (KRAS-down) epithelial state, metabolic and EMT / matrix programmes recur (Figure 3).
- **EMT states are not corners.** Only 14% of corners of significant fits are dominated (≥ 60%) by a hybrid state, and those are mostly cycling or MYC/growth corners (hybrid-state cells that divide); 14 of 19 EMT / matrix corners are mesenchymal or late-EMT, not hybrid.
- **So** a trade-off structure exists inside single tumours and lines, between proliferation, growth, differentiation and EMT / matrix programmes; EMT itself behaves like an axis running between epithelial and EMT / matrix corners, not like a set of task corners.

5. Question 2: are hybrid cells more often generalists than other cells?

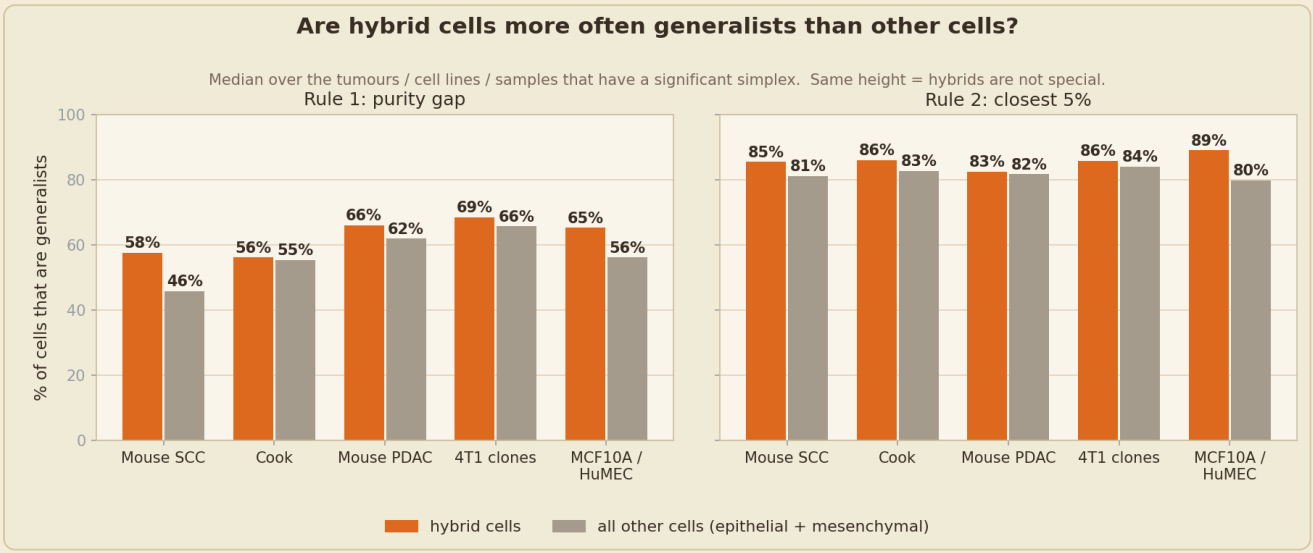


Figure 19. The main answer. For each dataset, orange = % of hybrid cells that are generalists, grey = % of all other cells that are generalists; bars are medians over the units with a significant simplex. Same height means hybrids are not special. Rule 3 is left out because it calls about 100% of every group generalist.

Medians over the units with a significant simplex (units without one have no meaningful corners, so they are left out of the medians but shown faded in the per-dataset figures).

dataset	gene set	units with a significant simplex	rule 1: hybrids vs others generalist	rule 2: hybrids vs others	verdict (rule 1)
Mouse SCC	Entire gene set	5	53% vs 41%	87% vs 81%	hybrids are clearly more often generalists
Mouse SCC	KS EMT genes	4	64% vs 49%	80% vs 82%	hybrids are clearly more often generalists
Mouse SCC	Hallmark EMT genes	4	57% vs 47%	87% vs 81%	hybrids are clearly more often generalists
Cook	Entire gene set	2	38% vs 36%	86% vs 83%	no real difference
Cook	KS EMT genes	2	33% vs 41%	84% vs 78%	hybrids are less often generalists (they sit at corners)
Cook	Hallmark EMT genes	4	59% vs 55%	86% vs 83%	no real difference
Mouse PDAC	Entire gene set	10	61% vs 55%	84% vs 83%	hybrids are slightly more often generalists
Mouse PDAC	KS EMT genes	3	75% vs 77%	80% vs 80%	hybrids are less often generalists (they sit at corners)
Mouse PDAC	Hallmark EMT genes	4	78% vs 73%	80% vs 79%	no real difference
4T1 clones	Entire gene set	2	62% vs 56%	90% vs 84%	hybrids are slightly more often generalists
4T1 clones	KS EMT genes	2	72% vs 73%	85% vs 79%	no real difference
4T1 clones	Hallmark EMT genes	1	94% vs 63%	94% vs 85%	hybrids are clearly more often generalists
MCF10A/HuMEC	Entire gene set	7	61% vs 60%	88% vs 80%	hybrids are clearly more often generalists
MCF10A/HuMEC	KS EMT genes	5	64% vs 55%	91% vs 80%	hybrids are slightly more often generalists
MCF10A/HuMEC	Hallmark EMT genes	6	69% vs 56%	90% vs 79%	hybrids are clearly more often generalists

5.1 Mouse SCC



Figure 20. Mouse SCC, every unit: orange = % of hybrid cells that are generalists, grey = % of other cells (rule 1); the number is orange minus grey in percentage points; faded = no significant simplex.

5.2 Cook

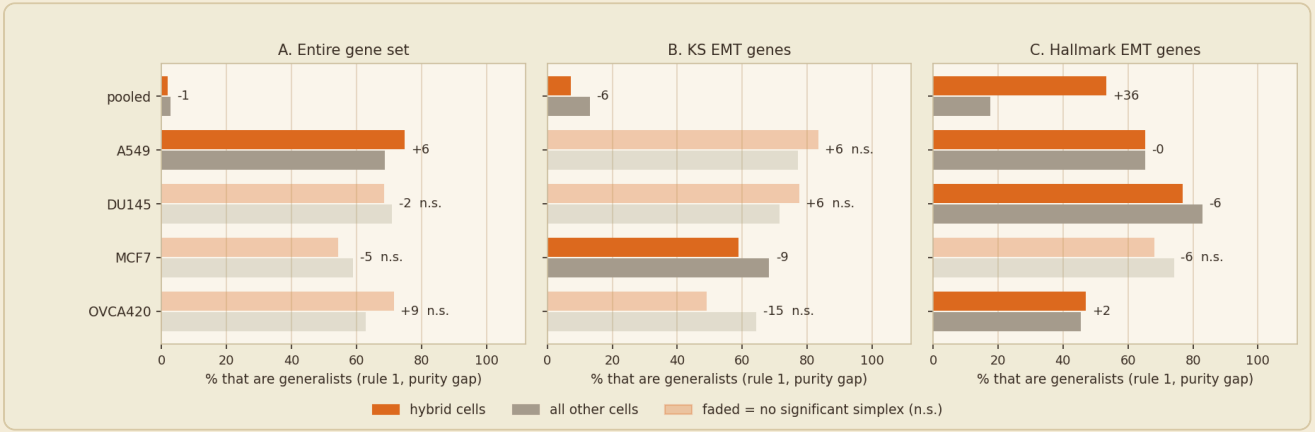


Figure 21. Cook, every unit: orange = % of hybrid cells that are generalists, grey = % of other cells (rule 1); the number is orange minus grey in percentage points; faded = no significant simplex.

5.3 Mouse PDAC



Figure 22. Mouse PDAC, every unit: orange = % of hybrid cells that are generalists, grey = % of other cells (rule 1); the number is orange minus grey in percentage points; faded = no significant simplex.

5.4 4T1 clones

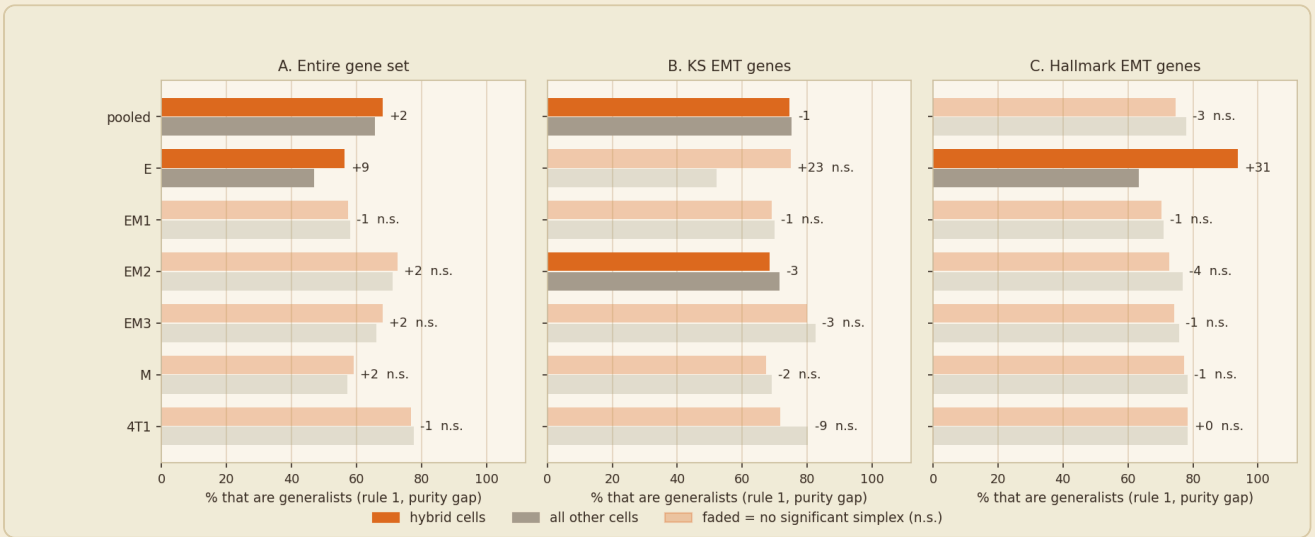


Figure 23. 4T1 clones, every unit: orange = % of hybrid cells that are generalists, grey = % of other cells (rule 1); the number is orange minus grey in percentage points; faded = no significant simplex.

5.5 MCF10A/HuMEC

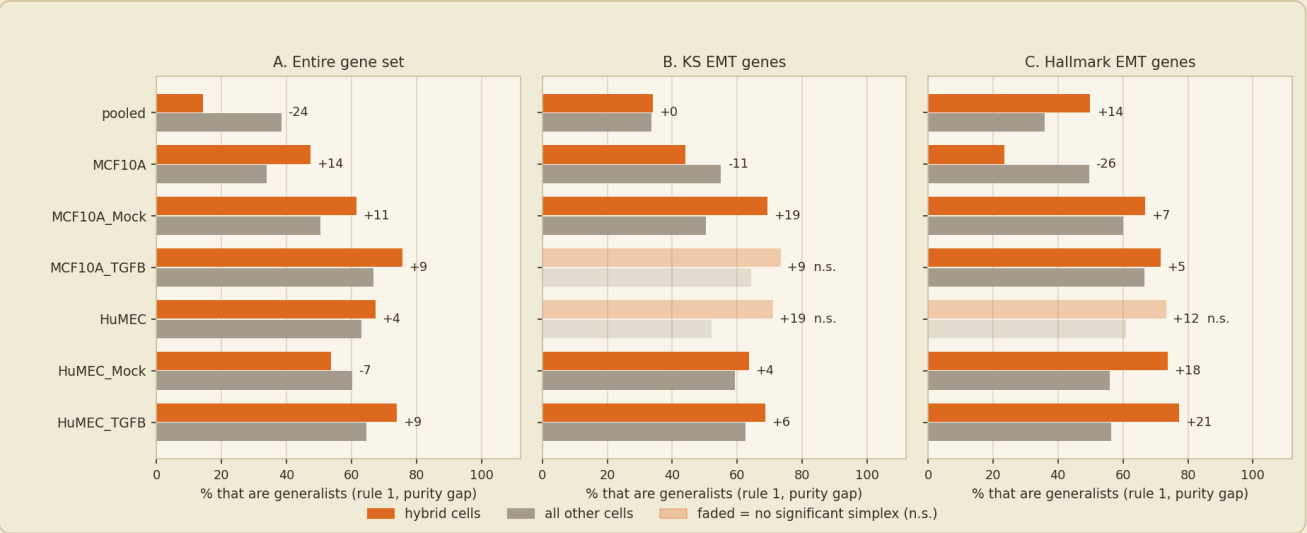


Figure 24. MCF10A/HuMEC, every unit: orange = % of hybrid cells that are generalists, grey = % of other cells (rule 1); the number is orange minus grey in percentage points; faded = no significant simplex.

5.6 What we saw (Question 2)

- **Most cells are generalists, hybrid or not**: rule 1 median 62% of hybrids vs 56% of other cells; rule 2 86% vs 81%; rule 3 about 100% for both. The rules differ by construction: rule 3 almost never calls a specialist, rule 2 lets only 5% of cells per corner be specialists, rule 1 is the one that separates groups.
- **Hybrids lean slightly toward the interior.** Median difference (hybrids minus other cells, rule 1) per dataset: Mouse SCC +10, MCF10A/HuMEC +6, Mouse PDAC +5, 4T1 clones +2, Cook +0 points; across all 61 significant fits +6 points. So the tilt is clearest in Mouse SCC and smallest in Cook.
- **Exceptions**: in 7 fits hybrids are much *less* often generalists than other cells (Mouse SCC SC2b3 (Entire gene set); Mouse PDAC M2 (Entire gene set); Mouse PDAC M2-Met (Entire gene set); Mouse PDAC M2 (KS EMT genes); MCF10A/HuMEC pooled (4 samples) (Entire gene set); MCF10A/HuMEC MCF10A (KS EMT genes); MCF10A/HuMEC MCF10A (Hallmark EMT genes)). In these fits hybrid-state cells sit at corners more often than other cells.
- **No hybrid corner.** In no dataset do hybrid cells form their own corner. This fits the picture of EMT as a continuum with hybrids as mixtures.

6. Caveats

- Many fits per dataset (gene sets, k); no multiplicity correction. p cannot go below about 0.005 (200 shuffles) or 0.01 (100 shuffles).
- Hybrid labels for PDAC, 4T1, GSE114687 and Cook are cut from the same KS score used for the KS-gene fits (partly circular). Cook's bins are pooled across lines, so a line's hybrid share reflects the line (A549 about 71%, MCF7 about 11%).

- Corner names are automatic (top Hallmark set); they summarise, and should be sanity-checked against the marker genes and the biology.
- Design limits: one tumour per clone (4T1), two mice (PDAC), no replicates (GSE114687), tumour / assay / state tangled (SCC). These results do not show that EMT is or is not a Pareto trade-off in tumours.
- Within-unit fits are capped at 4,000 random cells; Hallmark and cleaned-pooled fits were run only for k = 3 to 5.

7. Code and files

step	script	what it does
data	`01_build_anndata.py`, `make_clean_pooled.py`	expression matrices; cleaned copies for pooled cleaned fits
fits	`02_pca.py`, `03_fit_and_test.py`, `08_within_unit.py`, `07_within_line_clean.py`, `08d_within_tumour_geneset.py`	PCA, PCHA, permutation test (per dataset `codes/`)
queues	`pooled_queue.py`, `within_light_queue.py`	the Hallmark and missing fits run on 21 to 22 October
collect	`analyze.py`	all fits → best k per arm; three rules → `best_fits.csv`
main fits	`build_explainer_ppt.py`	main-fit policy → `main_fits.csv`; the slide deck
corners	`corners.py`	Hallmark naming of every corner → `corner_annotations.csv`
this report	`build_final_report.py`	figures in `figures/report/`; this document